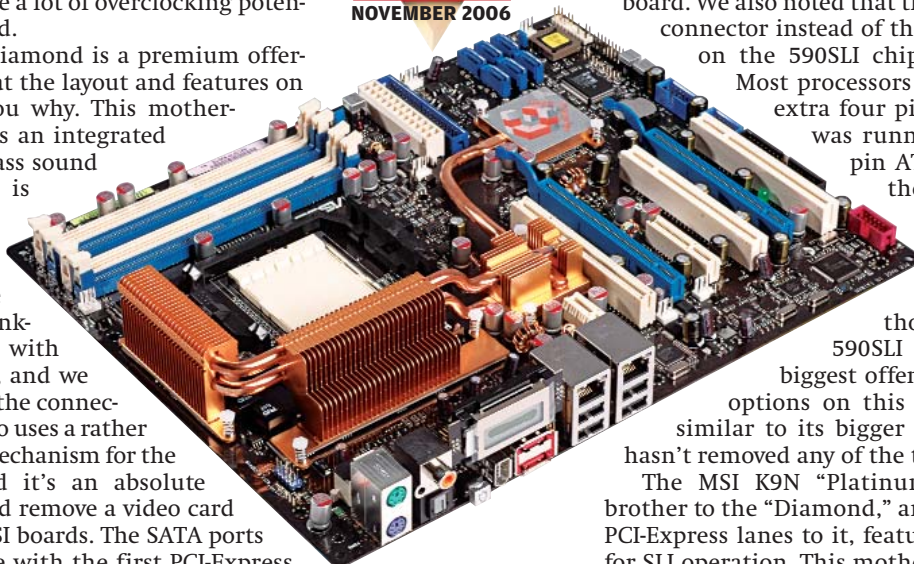


three PCI-Express slots, which look like x16 slots at first glance, but the one in the middle is an x4 slot. Due to the proximity, installing a graphics card in one slot renders the adjacent slot unusable, though this is a problem you'll see with most boards today. Gigabyte provides decent automatic overclocking support via the MB Intelligent Tweaker. The board frequency can be adjusted from 100 MHz all the way up to 500 MHz. Add to that the voltage tweaking available, and the control over multiplier ratios, and you have a lot of overclocking potential with this board.

The MSI K9N Diamond is a premium offering, and one look at the layout and features on offer will show you why. This motherboard also features an integrated Creative Audigy class sound solution, which is something you'll love if you're an audio enthusiast. We didn't have any of the heatsink-memory problems with the K9N Diamond, and we liked the layout of the connectors as well. MSI also uses a rather simple retention mechanism for the graphics slot, and it's an absolute breeze to insert and remove a video card from any of the MSI boards. The SATA ports are also not in-line with the first PCI-Express slot, which shows thoughtful design, because we've seen some longer cards like the GeForce 7900s, 7950s and Radeon X1800 and X1900 series actually interfere with the SATA ports. Though we noticed that MSI uses mostly electrolytic capacitors around the CPU region—not good for longevity—we found that it had the best layout amongst the high-end offerings.

The K9N Diamond features MSI's Dual "Core Cell," which equates to good automatic overclocking. The BIOS is by American Megatrends Inc. (AMI), and has enough options for decent



ASUS Crosshair
A combination of looks, features, and performance

manual overclocking. However, it's just not in the same league as the 590SLI boards from ASUS and Biostar as far as overclocking goes.

There were two AM2 motherboards based around NVIDIA's nForce 570SLI chipset, the Gigabyte GA-M57SLI-S4 and MSI's K9N Platinum.

The layout of the Gigabyte 570SLI is similar to that of its bigger brother. It doesn't sport a heatpipe solution but uses a smaller heatsink to cool the chipset. Gigabyte uses a lot of electrolyte capacitors around the CPU region of this board. We also noted that the CPU gets a 4-pin connector instead of the 8-pin connectors on the 590SLI chipset-based boards.

Most processors don't require the extra four pins, and our FX-62 was running happily on 4-pin ATX power even on the 8-pin boards.

Once again, the memory-CPU heatsink battle ensued,

though the Tforce 590SLI was by far the

biggest offender. Overclocking options on this board was pretty similar to its bigger brother; Gigabyte hasn't removed any of the tweaking options.

The MSI K9N "Platinum" plays smaller brother to the "Diamond," and also gives up 16 PCI-Express lanes to it, featuring two x8 lanes for SLI operation. This motherboard has a very clean and no-nonsense layout, no extra glitter, just an attractive aluminium heatsink with "MSI" etched on the centre in silver. There's a simple chrome-plated heatsink covering the mofsets around the I/O panel. The SATA and other ATA ports are thoughtfully placed, and adequate spacing between the memory banks means your RAM will get adequate cooling even while running gruelling applications. There is adequate space around the CPU socket, and you'll be surprised how much difference even a few millimetres here and there makes. MSI

How We Tested

We had a mammoth four test beds running for this month's motherboard shootout—two AMD-based test beds and two Intel-based. Note that for motherboards with onboard video, we didn't use a graphics card, and used the onboard solution instead.

The test beds are as listed below.

Windows XP with Service Pack 2 was our OS of choice. As always, the benchmarks are broadly divided into real-world and synthetic. Synthetic benchmarks represent applications that simulate conditions that stress test certain PC subsystems, and finally display a "score" that is usually numeric, and can be used

to quantifiably compare performance.

Real-world benchmarks are individual applications that, by their very nature, stress out certain components of the computer's subsystem. For example, games will severely stress the graphics, memory and CPU subsystems (in that order), while video encoding is processor-dependant.

The synthetic benchmarks were:

PC Mark 2005: A very standard benchmark today, PC Mark has a complete suite of tests designed to stress out individual components in turn. We used the latest version available.

SiSoft Sandra 2007 Engineer: Another industry-wide standard

THE TEST BEDS

	AMD 939 platform	AMD AM2 platform	Intel Core 2 Duo	Intel Pentium 4 / Pentium D platform
Processor	AMD 64 X2 4800+ @ 2.4 GHz	AMD 64 FX-62 @ 2.8 GHz	Core 2 Extreme @ 2.93 GHz	Pentium 4 EE @ 3.2 GHz
Memory	512 x 2 DDR400 @ 2-2-2-5	512 x 2 DDR2 800 @ 4-4-4-12	512 x 2 DDR2 800 @ 4-4-4-12	512 x 2 DDR2 800 @ 4-4-4-12
Graphics Card	XFX 7900GTX XXX Edition	XFX 7900GTX XXX Edition	XFX 7900GTX XXX Edition	XFX 7900GTX XXX Edition
HDD	Hitachi Deskstar T7K 250 SATA 2	Hitachi Deskstar T7K 250 SATA 2	Hitachi Deskstar T7K 250 SATA 2	Hitachi Deskstar T7K 250 SATA 2